The Binomial Theorem

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

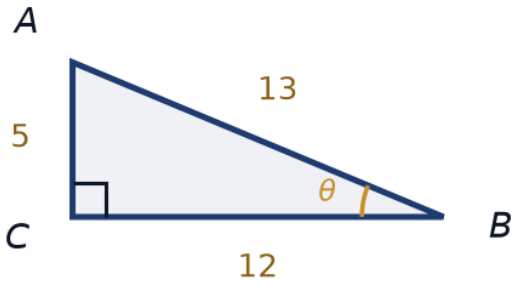
1 [Foundational] • Target: Binomial coefficients

Evaluate each binomial coefficient:

a $\binom{6}{2}$ **b** $\binom{8}{0}$ **c** $\binom{5}{5}$ **d** $\binom{10}{3}$

2 [Foundational] • Target: Binomial coefficients

Write down row 4 of Pascal's triangle (the row beginning $1, 4, \ldots$) and explain how it relates to the expansion of $(a+b)^4$.



3 [Foundational] • Target: Expansions and specific terms

Expand $(x+2)^4$ fully.

4 [Foundational] • Target: Expansions and specific terms

Find the coefficient of x^3 in the expansion of $(2x-1)^5$.

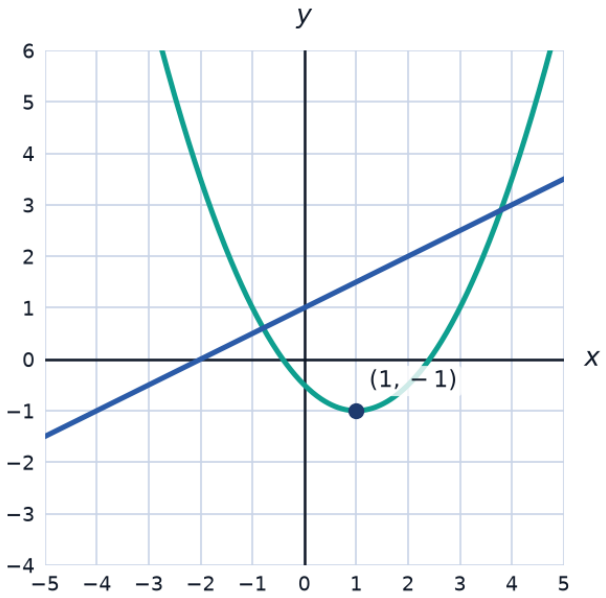
5 [Foundational] • Target: Expansions and specific terms

Find the term containing y^4 in the expansion of $(x+y)^7$.

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6 [Foundational] • Target: Expansions and specific terms

Use the binomial theorem to expand $(1 + 0.02)^5$ to the x^2-style second-order term, and hence estimate 1.02^5 to four decimal places.



7 [Foundational] • Target: More expansions and terms

Expand $(2x - y)^3$ fully.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More expansions and terms

Find the middle term of the expansion of $(x + 3)^6$ (the term with x^3).

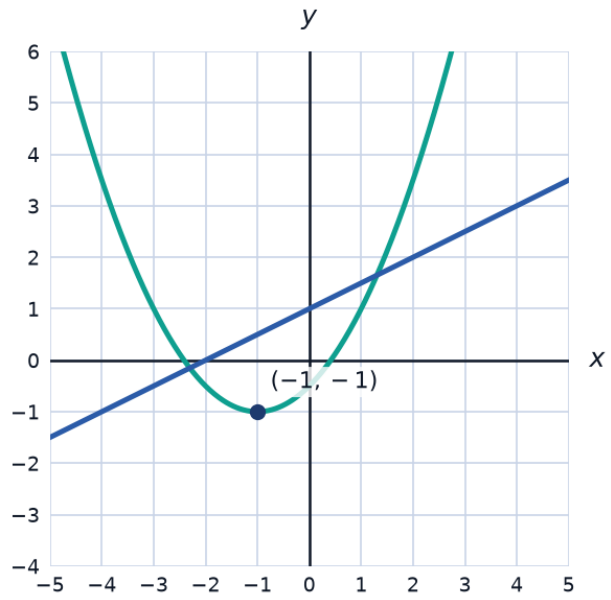
9 [Intermediate] • Target: More expansions and terms

Find the constant term in the expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

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10 [Intermediate] • Target: Sums of binomial coefficients

Evaluate $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$ without computing each term, using the identity $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.



11 [Intermediate] • Target: Sums of binomial coefficients

Use the fact that $\binom{n}{k} = \binom{n}{n-k}$ (symmetry of Pascal's triangle) to explain why $\binom{9}{2} = \binom{9}{7}$ without computing either value directly.

12 [Intermediate] • Target: Further expansions

Find the coefficient of x^4y^3 in the expansion of $(x + 2y)^7$.

13 [Intermediate] • Target: Further expansions

Find the first three terms in the expansion of $(1 - 2x)^6$ in ascending powers of x.

14 [Intermediate] • Target: Further expansions

Use the binomial theorem with $(1 - 0.01)^4$ to approximate 0.99^4 to four decimal places, using the first three terms.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

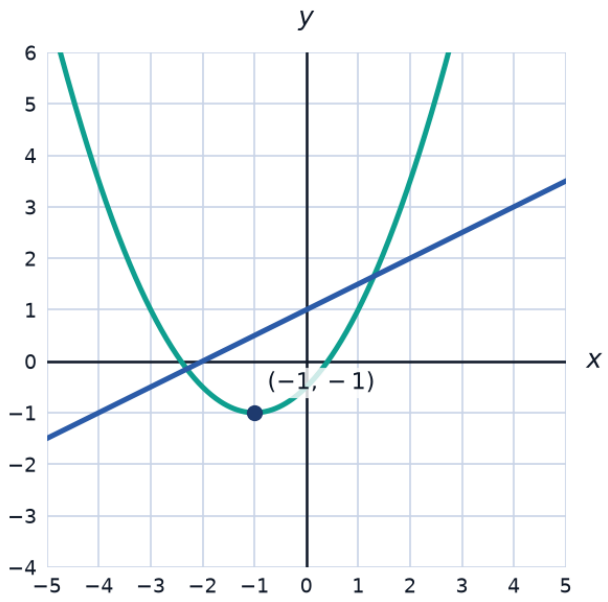
15 [Challenging] • Target: Deriving Pascal's triangle recursively • Cross-Lesson: 5.01 Vectors: Components and Operations

State Pascal's Rule, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, and use it to compute $\binom{6}{3}$ from row 5 of Pascal's triangle ($1, 5, 10, 10, 5, 1$).

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16 [Challenging] • Target: The general term formula • Cross-Lesson: 5.02 The Dot Product

State the general term (the $(k+1)$th term) formula for the expansion of $(a+b)^n$, and use it to find the 5th term of $(x-2)^8$ directly (without expanding the whole thing).



17 [Challenging] • Target: Combinatorial interpretation of binomial coefficients • Cross-Lesson: 5.03 Parametric Equations

Explain the combinatorial meaning of $\binom{n}{k}$, and use it to explain why $\binom{n}{0} = \binom{n}{n} = 1$ for any n.

18 [Challenging] • Target: Applying the binomial theorem to probability-style counting • Cross-Lesson: 6.01 Polar Coordinates and Conversions

A fair coin is flipped 5 times. Using $\binom{5}{k}$, find the number of outcomes with exactly 3 heads, and state the total number of possible outcomes (using 2^5).

19 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum of all the coefficients in the expansion of $(3x-2)^4$, using the fact that setting $x=1$ gives the sum of all coefficients.

20 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the coefficient of x^5 in the expansion of $(x^2+3)(x-1)^6$, by finding the relevant terms from $(x-1)^6$ first.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Pascal's Triangle and Combinations

State the formula for the combination $\binom{n}{r}$.

a. $\frac{n!}{r!\,(n-r)!}$

b. $\frac{n!}{(n-r)!}$

c. $\frac{n!}{r!}$

d. $n \times r$

22 [Foundational] • Target: Pascal's Triangle and Combinations

Evaluate $\binom{6}{2}$.

a. 15

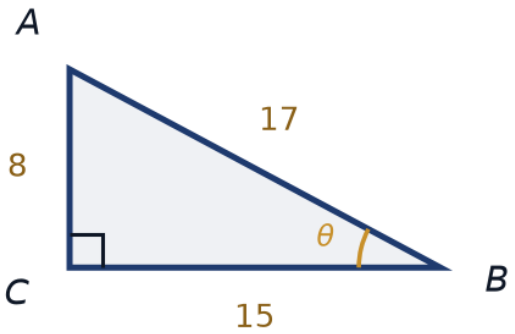
b. 30

c. 12

d. 20

23 [Foundational] • Target: Pascal's Triangle and Combinations

What is row 4 of Pascal's Triangle (corresponding to $n = 4$)?



a. 1, 4, 6, 4, 1

b. 1, 3, 3, 1

c. 1, 5, 10, 10, 5, 1

d. 1, 4, 8, 4, 1

24 [Foundational] • Target: Expanding Binomials

State the Binomial Theorem expansion for $(a + b)^n$.

a. $\sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$

b. $\sum_{r=0}^{n} a^{n-r} b^r$

c. $\sum_{r=1}^{n} \binom{n}{r} a^r b^{n-r}$

d. $\sum_{r=0}^{n} n!\, a^r b^r$

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25 [Foundational] • Target: Expanding Binomials

Expand $(x+2)^4$.

a. $x^4+8x^3+24x^2+32x+16$

b. $x^4+4x^3+6x^2+4x+16$

c. x^4+16

d. $x^4+8x^3+16x^2+32x+16$

26 [Foundational] • Target: Expanding Binomials

Find the first three terms in the expansion of $(2x-1)^5$ in descending powers of x.

a. $32x^5-80x^4+80x^3$

b. $32x^5-16x^4+8x^3$

c. $32x^5+80x^4+80x^3$

d. $32x^5-40x^4+20x^3$

27 [Foundational] • Target: Finding Specific Terms

State the formula for the $(r+1)$-th term in the expansion of $(a+b)^n$.

a. $T_{r+1}=\binom{n}{r}a^{n-r}b^r$

b. $T_{r+1}=\binom{n}{r+1}a^{n-r}b^r$

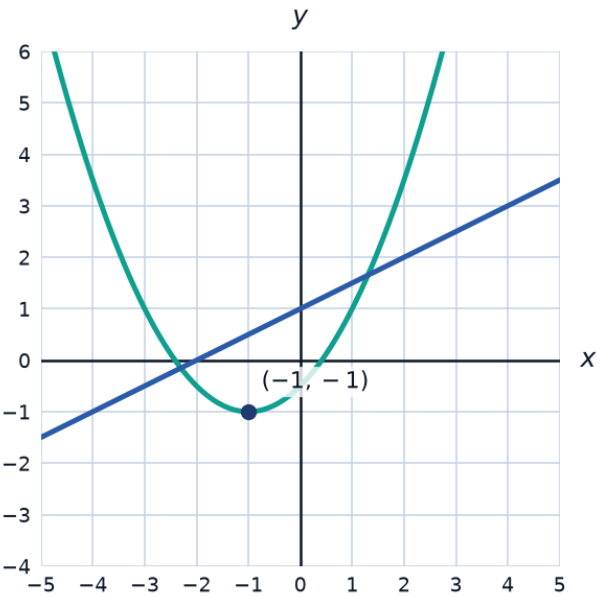
c. $T_{r+1}=\binom{n}{r}a^rb^{n-r}$

d. $T_{r+1}=a^{n-r}b^r$

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28 [Foundational] • Target: Finding Specific Terms

Find the 4th term in the expansion of $(x + 3)^7$.



a. $945x^4$

b. $315x^4$

c. $2835x^3$

d. $105x^5$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding Specific Terms

Find the coefficient of x^3 in the expansion of $(2x - 3)^6$.

a. -4320

b. 4320

c. -1440

d. 2160

30 [Intermediate] • Target: Term Independent of x and Sum of Coefficients

Find the term independent of x (the constant term) in the expansion of $\left(x + \frac{2}{x}\right)^6$.

a. 160

b. 80

c. 64

d. 20

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31 [Intermediate] • Target: Term Independent of x and Sum of Coefficients

What is the sum of all coefficients in the expansion of $(2x - 3y)^4$?

a. 1

b. -1

c. 625

d. 81

32 [Intermediate] • Target: Term Independent of x and Sum of Coefficients

What is the sum of the elements in row n of Pascal's Triangle (i.e. $\sum_{r=0}^{n} \binom{n}{r}$)?

a. 2^n

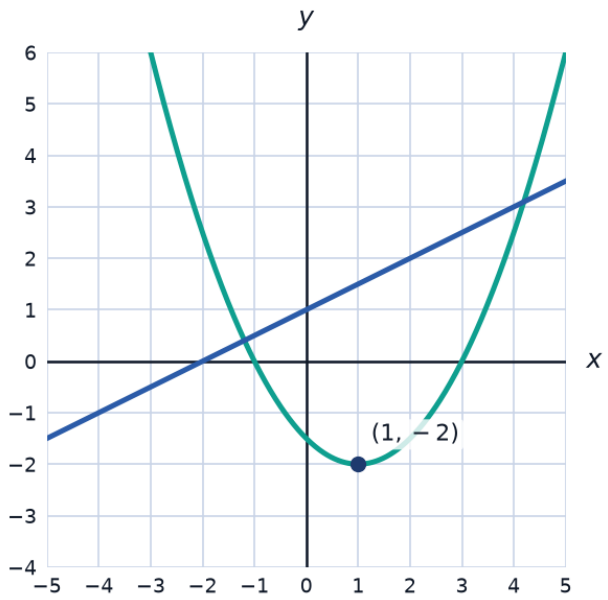
b. n^2

c. $2n$

d. 2^{n-1}

33 [Intermediate] • Target: Finding Specific Terms

A sequence is defined by $a_1 = 1$ and $a_{n+1} = a_n + 2n + 1$ for $n \geq 1$. Prove by induction that $a_n = n^2$ for all positive integers n. Then, consider the series $S_N = \sum_{n=1}^{N} a_n$. Find a closed-form expression for S_N and use it to determine the value of $\sum_{n=1}^{10} (a_n - a_{n-1})$ where $a_0 = 0$.



a. 100

b. 110

c. 10

d. 210

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34 [Intermediate] • Target: Term Independent of x and Sum of Coefficients

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of a_{2023}.

a. 2

b. 0

c. -2

d. 1

35 [Intermediate] • Target: Binomial coefficients

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -1$ and $d = 3$.

a. 128

b. 135

c. 115

d. 125

36 [Intermediate] • Target: Expansions and specific terms

Find the sum of the first 12 terms of the arithmetic series with $a_1 = -4$ and $d = 3$.

a. 138

b. 153

c. 150

d. 162

37 [Intermediate] • Target: More expansions and terms

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -5$ and $d = 5$.

a. 185

b. 175

c. 180

d. 165

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Sums of binomial coefficients • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -6$ and $d = 2$.

a. 32

b. 20

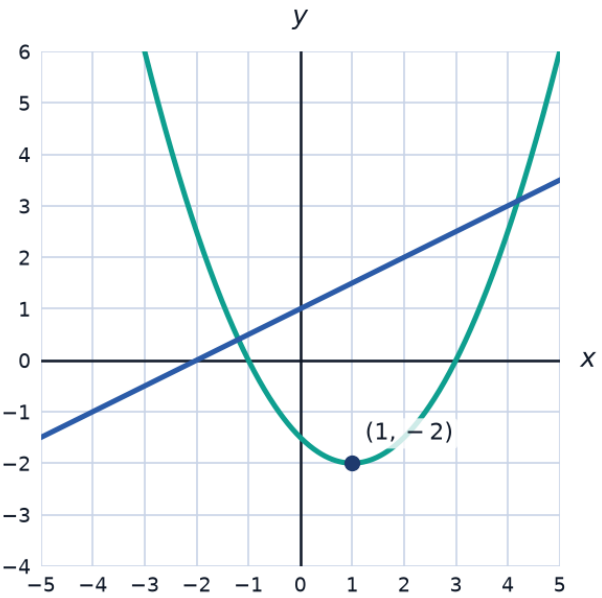
c. 30

d. 40

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39 [Challenging] • Target: Further expansions • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the sum of the first 12 terms of the arithmetic series with $a_1 = -1$ and $d = 1$.



a. 66

b. 42

c. 55

d. 54

40 [Challenging] • Target: Deriving Pascal's triangle recursively • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -4$ and $d = 2$.

a. 40

b. 50

c. 60

d. 52

41 [Challenging] • Target: The general term formula • Cross-Lesson: 7.02 Logarithms and Their Properties

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -1$ and $d = 2$.

a. 70

b. 90

c. 80

d. 82

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42 [Challenging] • Target: Combinatorial interpretation of binomial coefficients • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Find the sum of the first 12 terms of the arithmetic series with $a_1 = 0$ and $d = 4$.

a. 268

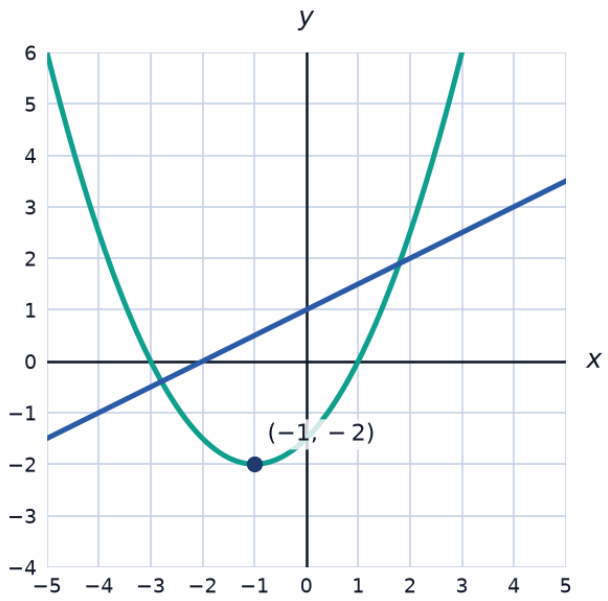
b. 264

c. 276

d. 252

43 [Challenging] • Target: Applying the binomial theorem to probability-style counting • Cross-Lesson: 7.04 Applications: Growth and Decay

Find the sum of the first 12 terms of the arithmetic series with $a_1 = 5$ and $d = 2$.



a. 192

b. 180

c. 204

d. 194

44 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find the sum of the first 10 terms of the arithmetic series with $a_1 = 1$ and $d = 1$.

a. 56

b. 55

c. 45

d. 65

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45 [Challenging] • Target: Pascal's Triangle and Combinations • Cross-Lesson: 8.02 Graphing Rational Functions

Find the sum of the first 10 terms of the arithmetic series with $a_1 = -3$ and $d = 1$.

a. 16

b. 15

c. 5

d. 25

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